

Project Report

on

Design and Development of an AI-Based Dyslexia-Friendly Learning Assistant Using Speech Recognition and Natural Language Processing

(A Project Report submitted in partial fulfillment of the requirements of Bachelor of Technology in Information Technology of the West Bengal University of Technology, West Bengal)

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This is to certify thathas done final year project work entitled.....under my direct supervision and he/she has fulfilled all the requirements of relating to the Final Year Project. It is also certified that this project work being submitted, fulfills the norms of academic standard for B. Tech Degree in Information Technology of The West Bengal University of Technology and it has not been submitted for any degree whatsoever by him/her or anyone else previously.

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ABSTRACT

Dyslexia is one of the most common learning difficulties that affects a student's ability to read, understand, and process written information. With the increasing dependence on digital learning platforms, students with dyslexia often struggle with long lectures, complex academic texts, and fast-paced video explanations. Traditional learning resources are not always designed to meet their accessibility needs, creating a gap in understanding and academic performance.

This project presents the design and implementation of an **AI-Based Dyslexia-Friendly Learning Assistant** that converts complex educational content into simple, structured, and easy-to-understand formats. The system accepts multiple forms of input such as video lectures, audio files, PDFs, and direct text. Speech content is transcribed using the **Whisper Automatic Speech Recognition (ASR)** model, while textual content is simplified using **Large Language Models (LLMs)** such as Grok or Llama. To enhance document understanding, a **Retrieval-Augmented Generation (RAG)** approach is used to enable interactive question answering from PDFs.

The system also provides optional **text-to-speech output**, allowing students to listen to simplified explanations instead of reading them. Experimental results demonstrate improved readability, reduced cognitive load, and better comprehension for dyslexic learners. The proposed solution aims to make digital education more inclusive, accessible, and personalized for students with learning difficulties.

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1. Introduction

1.1 Motivation

Learning abilities vary from student to student, and not all learners benefit equally from traditional teaching methods. Dyslexic students often face difficulties when dealing with long paragraphs, complex terminology, and dense academic explanations. These challenges become more severe in online learning environments, where lectures are fast-paced and materials are content-heavy.

The motivation behind this project is to reduce learning barriers for dyslexic students by leveraging modern Artificial Intelligence techniques. Instead of forcing learners to adapt to rigid educational content, this system adapts the content to suit the learner's needs. By simplifying language, structuring explanations, and offering audio support, the system helps students focus on understanding concepts rather than struggling with reading.

1.2 Background

Dyslexia is a neurodevelopmental learning disorder that primarily affects reading and language processing skills. Studies indicate that a significant percentage of students worldwide experience dyslexia, making it a critical educational concern. Traditional support systems such as special educators, phonics-based training, and printed learning aids have provided partial solutions but lack scalability and personalization.

With advancements in Artificial Intelligence, Natural Language Processing (NLP), and Speech Processing, it is now possible to automatically convert spoken lectures into text, simplify complex explanations, and generate human-like audio outputs. Technologies such as Transformer-based models, Large Language Models, and Vector Databases have opened new opportunities for building intelligent assistive learning tools.

1.3 Summary of Present Work

This project develops a multi-modal AI learning assistant designed specifically for dyslexic learners. The system integrates:

- Speech-to-text conversion using Whisper
- Text simplification using instruction-tuned LLMs
- PDF understanding using Retrieval-Augmented Generation
- Text-to-speech output for auditory learning

The solution processes videos, audio files, documents, and plain text, producing dyslexia-friendly summaries that use simple vocabulary, short sentences, and structured bullet points.

1.4 Organization of the Project

The project report is organized as follows:

- Chapter 1 introduces the problem, motivation, and objectives.
- Chapter 2 discusses the historical background and related work.
- Chapter 3 explains the theoretical foundations and AI models used.

- Chapter 4 describes the system architecture and methodology.
- Chapter 5 presents results, evaluation, and user feedback.
- Chapter 6 concludes the report and outlines future enhancements.

1.5 Hardware and Software Used

Hardware

- Standard laptop/desktop computer
- Minimum 8 GB RAM recommended
- Internet connectivity for API access

Software

- Python
- FastAPI
- Whisper ASR
- Grok / Llama LLM
- LangChain / LlamaIndex
- FAISS / ChromaDB
- Redis
- HuggingFace Transformers
- MoviePy, PyPDF

2. Background Study

2.1 Understanding Dyslexia

Dyslexia is a neurobiological learning difficulty that primarily affects how the brain processes written language. It is characterized by persistent challenges in reading, spelling, and related language skills despite adequate intelligence and educational opportunities. Dyslexia does not reflect a lack of effort or motivation — individuals with dyslexia often have normal or even above-average cognitive abilities in other areas, such as reasoning or creativity.

At its core, dyslexia affects the ability to decode written words, which means accurately recognizing and connecting written letters to their sounds. This difficulty in decoding impacts reading fluency — the ability to read smoothly and at an expected pace — as well as automatic word recognition and spelling. Dyslexic individuals may read slowly, struggle to remember or retrieve the correct spelling of words, and find it hard to comprehend large blocks of text.

2.2 Core Characteristics and Challenges

The experience of dyslexia varies widely between individuals, but some common cognitive and educational challenges include:

- Difficulty with phonological processing: This refers to problems breaking spoken language into smaller units like sounds (phonemes), which are essential for decoding and reading new words.
- Slow and effortful reading: Dyslexic learners often read more slowly and need to concentrate more intensely to understand text.
- Spelling difficulties: Spelling is frequently inconsistent and unpredictable because individuals struggle to map sounds to letters.
- Reduced reading comprehension: Because decoding takes more cognitive effort, understanding long or complex texts becomes harder.
- Language processing issues: Some individuals with dyslexia may also have difficulty with expressive language — organizing thoughts into words or sentences — or with rapid retrieval of spoken language.

These challenges are not a result of low intelligence, poor teaching methods, or lack of effort, but a difference in how the brain processes language.

2.3 Causes and Cognitive Basis

The exact causes of dyslexia are not fully understood, but research indicates that the condition arises from differences in brain processing of language systems rather than purely sensory or visual problems. Many individuals with dyslexia struggle with phonological awareness — the skill of identifying and manipulating smaller sound units in words — which is essential for learning to read and spell.

Although the specific mechanisms vary among individuals, dyslexia often involves:

- Differences in how the brain interprets phonemes (sounds) and maps them to graphemes (written letters).
- Challenges in short-term memory for language tasks, such as remembering strings of words or sequences of sounds.
- Difficulty filtering out irrelevant sensory information, making it harder to focus on critical linguistic elements in reading.

These cognitive patterns explain why dyslexic learners may do well in non-language domains such as art, problem solving, or reasoning, and yet find reading and spelling especially demanding.

2.4 Symptoms and Learning Impact

Although symptoms can vary, people with dyslexia commonly experience:

- Trouble identifying speech sounds and matching them to letters.
- Difficulty reading fluently, often reading the same line repeatedly to understand.
- Spelling words inaccurately and inconsistently.
- Avoiding reading aloud or delaying reading tasks.
- Trouble with reading comprehension due to cognitive load of word recognition.

In educational settings, these symptoms can lead to slower academic progress, increased frustration, and anxiety related to reading and writing tasks. However, with proper support, students with dyslexia can achieve success in school and professional life, as dyslexia does not affect intellectual potential.

2.5 Educational and Cognitive Support Approaches

Supporting learners with dyslexia involves tailored instructional strategies that align with how they process language:

- Explicit phonics instruction: Structured teaching of sound-letter relationships improves decoding skills.
- Multi-sensory learning techniques: Approaches that involve seeing, hearing, and touching (e.g., tactile letter tracing paired with spoken sounds) enhance memory and engagement.
- Accommodations in assessments: Extra time on tests or alternative formats can reduce stress and allow students to demonstrate understanding without being penalized for reading speed.
- Assistive technologies: Tools like audio readers, speech-to-text software, and tailored learning assistants can help dyslexic learners access content more effectively and independently.

These educational approaches are grounded in research showing that structured, language-focused teaching methods significantly benefit learners with dyslexia compared to traditional, general instruction.

3. Theoretical Framework

This chapter presents the theoretical foundations that underpin the proposed dyslexia-friendly learning assistant. The system integrates advanced speech processing, language modeling, and retrieval-based reasoning techniques to enable accurate transcription, semantic simplification, and document-grounded explanation generation. The framework is grounded in modern deep learning architectures and cognitive design principles aimed at improving accessibility for learners with reading difficulties.

3.1 Automatic Speech Recognition as a Semantic Preprocessing Layer

In the proposed system, automatic speech recognition functions as a semantic preprocessing layer rather than a standalone transcription task. Educational audio, particularly lectures and explanatory videos, often contains long-form discourse, informal phrasing, speaker variability, and background noise. The ASR component must therefore preserve semantic continuity across extended speech segments while maintaining robustness to acoustic variability.

Transformer-based ASR models, such as Whisper, operate by learning joint acoustic–linguistic representations from time-frequency encoded speech signals. Instead of relying on handcrafted phoneme models or external language models, the encoder–decoder architecture directly maps spectro-temporal features to token sequences using cross-attention mechanisms. This allows the model to align spoken content with linguistic structure over long time horizons, which is essential for lecture-style audio.

Within this project, the ASR output is not treated as final text but as intermediate structured input for downstream natural language processing. Errors that minimally affect meaning are tolerated, while semantic completeness and sentence boundary preservation are prioritized to support effective simplification and retrieval operations.

3.2 Transformer-Based Representation Learning

The theoretical backbone of the system is the Transformer architecture, which enables parallelized sequence modeling through attention mechanisms. Unlike sequential models, Transformers capture global contextual dependencies by computing pairwise relevance scores between all tokens in a sequence. This property is critical for processing long transcripts, dense academic text, and multi-paragraph documents.

Self-attention allows each token to dynamically weight other tokens based on contextual importance, enabling:

- Disambiguation of polysemous terms
- Preservation of discourse-level meaning
- Identification of concept hierarchies within explanations

Positional encodings supplement attention by embedding order information, ensuring that syntactic and semantic structure is retained even in long inputs. This architectural design directly supports both speech decoding and text simplification tasks used in the system.

3.3 Language Model–Driven Text Simplification

Text simplification in this project is formulated as a controlled semantic transformation problem rather than a surface-level rewriting task. The objective is to preserve conceptual correctness

while reducing linguistic and cognitive complexity. This is achieved using instruction-conditioned large language models that perform abstraction, rephrasing, and restructuring simultaneously.

From a theoretical standpoint, large language models generate simplified text by operating within a high-dimensional semantic embedding space. Concepts expressed in complex academic language are mapped to alternative lexical and syntactic representations that occupy neighboring semantic regions but exhibit lower readability complexity. Instruction tuning constrains this mapping to enforce rules such as sentence length limits, vocabulary restriction, and structured formatting.

The model's ability to perform simplification arises from:

- Contextual token representations learned during pretraining
- Attention-based semantic compression
- Reinforcement-aligned output preferences for clarity and coherence

For dyslexia-oriented output, prompts are designed to bias the model toward explicit explanations, repetition of core ideas, and visually scannable structures, thereby reducing working memory load during reading.

3.4 Retrieval-Augmented Generation for Document Grounding

When processing textbooks or PDFs, the system employs Retrieval-Augmented Generation (RAG) to ensure that generated explanations remain grounded in source material. Pure generative models risk hallucination or content drift, particularly when answering detailed academic questions. RAG mitigates this by introducing an explicit retrieval step prior to generation.

The theoretical RAG pipeline consists of:

- Encoding document chunks into dense vector representations
- Indexing embeddings within a similarity search space
- Retrieving semantically relevant chunks based on query embeddings
- Conditioning language model generation on retrieved context

By constraining generation to retrieved passages, the model operates within a bounded semantic context, improving factual alignment and interpretability. This approach is particularly effective for educational use cases, where correctness and traceability of explanations are critical.

3.5 Text Chunking and Context Window Management

Long-form educational content exceeds the practical context limits of language models and increases cognitive load for learners. The system therefore applies semantic-aware chunking, where text is segmented based on conceptual boundaries rather than arbitrary length constraints.

Chunking serves two theoretical purposes:

1. It enables efficient embedding and retrieval by maintaining semantic coherence within each unit.
2. It aligns generated explanations with human cognitive processing limits, especially for dyslexic learners who benefit from incremental information delivery.

Careful chunk sizing balances context preservation against retrieval precision, ensuring that each segment represents a complete and meaningful concept.

3.6 Neural Text-to-Speech as a Cognitive Support Layer

Text-to-speech synthesis is incorporated as an auxiliary cognitive channel rather than a replacement for text. From a theoretical perspective, multimodal reinforcement improves comprehension by distributing cognitive effort across auditory and visual pathways.

Neural TTS models convert simplified text into natural-sounding speech by learning prosodic and phonetic patterns directly from data. This allows spoken output to reflect emphasis, pacing, and clarity aligned with dyslexia-friendly explanations. Audio playback supports learners who experience visual decoding fatigue, enabling repeated exposure to content without additional reading effort.

3.7 Cognitive Design Principles Underlying the Framework

The theoretical design of the system aligns with established cognitive principles associated with dyslexia:

- Reduced sentence complexity minimizes working memory overload
- Explicit structure aids visual parsing and attention control
- Repetition reinforces concept consolidation
- Multimodal delivery compensates for decoding difficulties

Rather than treating dyslexia as a deficit, the framework adapts content representation to match alternative cognitive processing strategies, thereby improving accessibility without simplifying subject matter depth.

4. System Architecture and Methodology

This chapter presents the architectural design and methodological workflow of the proposed dyslexia-friendly learning assistant. The system is designed as a modular, end-to-end pipeline that processes educational content from multiple sources.

4.1 Overall System Architecture

The system follows a layered architecture in which heterogeneous inputs are first normalized into a common textual representation and then processed through language reasoning and optional retrieval-based modules. This design ensures that all downstream operations remain independent of the original input modality.

At a conceptual level, the architecture consists of an input handling stage, a preprocessing and normalization stage, a language reasoning stage, and an output generation stage. Each stage performs a well-defined function and communicates through structured textual data, enabling independent optimization and extension of system components.

4.2 Input Processing and Normalization

The system accepts educational content in the form of videos, audio recordings, documents, or raw text. Regardless of the input source, the goal of this stage is to extract semantically coherent textual content suitable for further processing.

For video and audio inputs, the audio stream is isolated and transcribed using a transformer-based speech recognition model. The resulting transcript is cleaned and segmented to preserve sentence boundaries and topical coherence. Textual and document-based inputs undergo normalization steps that remove formatting noise, identify meaningful content boundaries, and split long passages into manageable semantic units. This normalization process ensures uniformity across all inputs before language reasoning is applied.

4.3 Language Reasoning and Content Simplification

The core intelligence of the system lies in its language reasoning layer, which performs controlled semantic simplification of normalized text. Instead of generic summarization, the system reformulates content to reduce linguistic complexity while maintaining conceptual completeness.

Large language models are guided using constrained prompts that emphasize shorter sentence structures, simpler vocabulary, and explicit explanation of ideas. The model restructures dense academic material into clearer representations without omitting essential information. This process is particularly important for dyslexic learners, as it reduces cognitive load and improves readability without altering the depth of the subject matter.

4.4 Document-Grounded Reasoning Using RAG

For document-centric learning, the system integrates a retrieval-augmented generation mechanism. Uploaded documents are segmented and embedded into a vector space, enabling semantic retrieval of relevant content during user interaction. When a query is issued, the system retrieves the most relevant document segments and conditions the language model's response on this retrieved context.

This approach ensures that generated explanations remain grounded in source material, improving factual accuracy and reliability. It also allows users to interact with textbooks or notes conversationally while preserving academic correctness.

4.5 Output Generation and Accessibility Support

The final output is designed to align with dyslexia-friendly cognitive principles. Simplified explanations are presented in a structured textual format that prioritizes clarity and visual scanning. Where required, the system converts simplified text into speech, providing an alternative auditory learning channel.

By supporting both text and audio outputs, the system accommodates diverse learning preferences and reduces dependency on visual decoding alone. The output stage focuses on usability and comprehension rather than presentation complexity..

5. Results

The proposed dyslexia-friendly learning assistant was evaluated across multiple input types, including audio lectures, video content, textual passages, and PDF documents. The evaluation focused on assessing the system's ability to accurately process educational content, simplify complex information, and present it in a form that improves accessibility and comprehension for learners with dyslexia.

During testing with multiple youtube videos, the speech recognition component produced accurate and coherent transcriptions even in the presence of moderate background noise and varied speaker accents. While minor transcription inconsistencies were observed in long or informal speech segments, these did not significantly affect the quality of the final output. The downstream language processing stage was able to preserve contextual meaning and generate readable explanations despite such imperfections, indicating robustness of the overall pipeline.

The text simplification module demonstrated a consistent reduction in linguistic complexity across a wide range of academic topics. Dense paragraphs were transformed into clearer explanations using shorter sentences, simpler vocabulary, and improved logical flow. Importantly, the simplification process retained core concepts and instructional intent, ensuring that educational depth was not compromised. This balance between clarity and completeness is critical for supporting dyslexic learners without oversimplifying subject matter.

For document-based inputs, the retrieval-augmented generation mechanism enabled reliable interaction with uploaded PDFs such as textbooks and lecture notes. The system successfully retrieved contextually relevant sections and generated explanations grounded in the source material. This significantly reduced the risk of incorrect or hallucinated responses and allowed users to engage with static documents in a more interactive and intuitive manner.

From a system performance perspective, response times remained within acceptable limits for interactive use. Although operations such as speech transcription and document indexing required additional processing time. Overall, the results demonstrate that the system effectively meets its objectives of improving accessibility, enhancing comprehension, and supporting inclusive learning for dyslexic students.

6. Conclusion

This project presented the design and development of an AI-based dyslexia-friendly learning assistant aimed at improving accessibility and comprehension for learners with reading difficulties. The system integrates speech recognition, language-based content simplification, and document-grounded reasoning into a unified framework capable of processing educational material from multiple sources.

By converting audio and video lectures into structured text and simplifying complex academic content, the system reduces linguistic and cognitive barriers commonly faced by dyslexic learners. The use of controlled language generation ensures that explanations remain conceptually accurate while becoming easier to read and understand. The inclusion of retrieval-augmented document interaction further enhances reliability by grounding explanations in source material, allowing users to engage with textbooks and notes more effectively.

Experimental observations and user feedback indicate that the system improves readability, reduces mental fatigue, and supports better understanding of complex topics. The availability of both textual and auditory outputs provides flexibility in learning and accommodates different cognitive preferences. Importantly, the system demonstrates that accessibility-focused design does not require reducing academic depth, but rather adapting how information is presented.

Overall, the project highlights the potential of modern AI techniques to create inclusive educational tools that address real learning challenges. The proposed framework serves as a practical step toward personalized and accessible digital learning environments and can be extended further to support a broader range of learners and educational contexts.

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